Classifier ablation (Tier 1, item 19)

Does the bottleneck classifier contribute anything to the analytical roofline formula, or would any weighting of its three terms do as well?

MAPE % on `scaling_factor_p50`, multi-core compute signal, classifier refit inside every fold. `bw_only` is identical to the naive-linear baseline by construction. `oracle` picks the best of the three terms per row with hindsight, so it bounds what *any* weighting scheme could achieve.

weighting	LOMO	LOQO	LOMQO
learned	21.61	21.66	21.64
onehot	21.62	21.62	21.62
uniform	21.55	21.55	21.55
bw_only	23.05	23.05	23.05
cmp_only	20.06	20.06	20.06
oracle	17.07	17.07	17.07

Verdict

Under LOMO the learned classifier scores **21.61%** against **21.55%** for a fixed uniform weighting that uses no query-specific information at all — a difference of **-0.06pp**.

The classifier is decorative. A constant weighting matches it to within half a percentage point, so essentially all of the analytical formula's accuracy comes from the hardware ratios, not from knowing which resource a query is bound by. The component whose ground truth is 21 hand-assigned judgement calls is contributing nothing measurable.

The hand labels used as a hard one-hot score **21.62%**, so even perfect knowledge of the (hand-assigned) class is worth +1.43pp against the pure bandwidth ratio.

The per-row oracle bound is **17.07%**. That is the ceiling for any scheme that weights these three terms, so it also bounds how much a better classifier — or the measured elasticity vectors from item 17/18 — could possibly gain within this functional form. Beating it requires a different formula, not better weights.

The result that matters: `1/compute_ratio` alone is the best predictor

`cmp_only` — a single number, the ratio of multi-core aggregate GFLOPS, with no classifier, no bottleneck weighting and no second hardware axis — scores **20.06% MAPE**, and it is also the best on MdAPE (15.87%) and on median q-error. It beats the full analytical roofline (21.61%) and the bandwidth baseline (23.05%).

But the improvement is not statistically significant, and the weaker one is. Paired cluster bootstrap over the 5 held-out machines:

comparison	point	95% CI	verdict
naive-linear – analytical	+1.44pp	[0.32, 2.56]	**SIGNIFICANT**
naive-linear – cmp_only	+2.99pp	[-0.09, 6.07]	not significant
analytical – cmp_only	+1.55pp	[-0.41, 3.51]	not significant
gated – cmp_only	+2.25pp	[-0.39, 4.89]	not significant

This is not a contradiction, it is the whole lesson of a 5-machine evaluation. The analytical formula's small gain is *consistent* — it improves on the bandwidth ratio on nearly every fold, so the paired difference has low variance and the CI clears zero. `cmp_only`'s larger gain is *inconsistent*:

held-out machine	naive-linear	analytical	cmp_only
c5a	19.68	19.80	**20.42**
c7i	27.75	25.39	**21.83**
m5a	20.58	20.36	**21.67**
r5n	25.97	22.64	**18.32**
z1d	21.26	19.87	**18.08**

It wins big on r5n and z1d and loses on c5a and m5a. With 5 clusters that spread swamps the mean. **The honest statement is: `cmp_only` has the best point estimate on every error metric, and 5 machines are not enough to establish it.** Settling this is precisely what the Tier 2 dataset expansion is for.

Why this undermines the bottleneck taxonomy

bottleneck class	naive-linear	analytical	cmp_only
bandwidth	17.64	17.61	**14.10**
compute	22.63	17.55	**17.50**
io	28.79	28.76	**26.14**
mixed	23.04	21.58	**20.79**

`cmp_only` is the best predictor on **every** class, including the ones labelled `bandwidth` (14.10% vs 17.61%) and `io` (26.14% vs 28.76%). A query hand-labelled bandwidth-bound is predicted better by the compute ratio than by the bandwidth ratio. That is hard to reconcile with the labels describing a real property of the queries, and it is consistent with the classifier ablation above: the taxonomy is not carrying information.

One guard against over-reading this: the same `1/compute_ratio` predictor on the **single-core** compute signal scores **51.44%**, far worse than everything. So the finding is not that compute ratios are magic — it is that the *multi-core aggregate* measurement captures what differentiates these machines, which is the parallelism mechanism already confirmed causally (r = 0.458, p = 0.037).